

- Description

This module manages a smart water purifier system, ensuring that:

1. The total dissolved solids (TDS) level in the water is continuously monitored (in ppm).
2. If the TDS level exceeds a predefined maximum threshold, the system automatically activates a filter cleaning cycle to reduce the TDS.
3. If the TDS level is below the minimum threshold, the system remains idle to prolong filter life.

This system helps maintain optimal water quality by ensuring TDS levels are within safe limits and triggering filter maintenance when necessary.

- Specification

module SmartWaterPurifierControl

var currentTDS, maxTDSThreshold, minTDSThreshold: real

var filterCleaningActive: boolean

process updateTDS(newTDS: real) returns (currentTDS: real, filterCleaningActive: boolean)

functional scenario form:

currentTDS > maxTDSThreshold && filterCleaningActive = true, Filter cleaning activated.

||

currentTDS < minTDSThreshold && filterCleaningActive = false, Filter cleaning deactivated.

||

currentTDS <= maxTDSThreshold && filterCleaningActive = false, No action required.

end process

end module

- Java Code

```
package code;
public class c84_SmartWaterPurifier {
    private double maxTDSThreshold;
    private double minTDSThreshold;
    private double currentTDS;
    private boolean filterCleaningActive;
    public c84_SmartWaterPurifier(double minTDSThreshold, double maxTDS
Threshold, double initialTDS) {
        this.minTDSThreshold = minTDSThreshold;
        this.maxTDSThreshold = maxTDSThreshold;
        this.currentTDS = initialTDS;
        this.filterCleaningActive = false;
    }
}
```

```

    public void updateTDS(double newTDS) {
        currentTDS = newTDS;
        adjustFilterActivation();
        System.out.printf("Current TDS: %.2f ppm, Filter Cleaning: %s\n",
                           currentTDS, filterCleaningActive ? "ACTIVE" :
                           "INACTIVE");
        if (filterCleaningActive) {
            System.out.println("ALERT: TDS level high! Initiating filter cleaning cycle.");
        } else {
            System.out.println("TDS level is within acceptable limits. No action required.");
        }
    }

    private void adjustFilterActivation() {
        if (currentTDS > maxTDSThreshold) {
            filterCleaningActive = true;
        } else if (currentTDS < minTDSThreshold) {
            filterCleaningActive = false;
        } else {
            filterCleaningActive = false;
        }
    }

    public double getCurrentTDS() {
        return currentTDS;
    }

    public boolean isFilterCleaningActive() {
        return filterCleaningActive;
    }

    public static void main(String[] args) {
        c84_SmartWaterPurifier purifier = new c84_SmartWaterPurifier(200.0, 400.0, 300.0);

        // testcase-VT:
        //     purifier.updateTDS(350.0);
        //     purifier.updateTDS(450.0);
        //     purifier.updateTDS(180.0);

        // testcase-FT:
        //     purifier.updateTDS(1.7976931348623157E308);
        //     purifier.updateTDS(-1.7976931348623157E308);
        //     purifier.updateTDS(0.0);

        // testcase-Z3:

```

```
//      purifier.updateTDS(350.0);
//      purifier.updateTDS(450.0);
//      purifier.updateTDS(180.0);

// testcase-UVT:
purifier.updateTDS(350.0);
purifier.updateTDS(450.0);
purifier.updateTDS(180.0);
}
}
```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c84_SmartWaterPurifier.java	100.0 %	8	0	8
▼ c84_SmartWaterPurifier	100.0 %	8	0	8
main(String[])		0	0	0
c84_SmartWaterPurifier(double,		0	0	0
adjustFilterActivation()	100.0 %	4	0	4
updateTDS(double)	100.0 %	4	0	4

3. Vibration testing instruction coverage: 93.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	93	21,109	21,202
▼ src	0.4 %	93	21,109	21,202
▼ code	0.4 %	93	21,109	21,202
▼ c84_SmartWaterPurifier.java	93.9 %	93	6	99
▼ c84_SmartWaterPurifier	93.9 %	93	6	99
main(String[])	100.0 %	17	0	17
c84_SmartWaterPurifier(double,	100.0 %	15	0	15
adjustFilterActivation()	100.0 %	24	0	24
updateTDS(double)	100.0 %	37	0	37

4. Fuzz testing branch coverage: 87.5%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.8 %	7	835	842
▼ src	0.8 %	7	835	842
▼ code	0.8 %	7	835	842
▼ c84_SmartWaterPurifier.java	87.5 %	7	1	8
▼ c84_SmartWaterPurifier	87.5 %	7	1	8
main(String[])		0	0	0
c84_SmartWaterPurifier(double,		0	0	0
adjustFilterActivation()	75.0 %	3	1	4
updateTDS(double)	100.0 %	4	0	4

5. Fuzz testing instruction coverage: 90.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	90	21,112	21,202
▼ src	0.4 %	90	21,112	21,202
▼ code	0.4 %	90	21,112	21,202
▼ c84_SmartWaterPurifier.java	90.9 %	90	9	99
▼ c84_SmartWaterPurifier	90.9 %	90	9	99
main(String[])	100.0 %	17	0	17
c84_SmartWaterPurifier(double,	100.0 %	15	0	15
adjustFilterActivation()	87.5 %	21	3	24
updateTDS(double)	100.0 %	37	0	37

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c84_SmartWaterPurifier.java	100.0 %	8	0	8
▼ c84_SmartWaterPurifier	100.0 %	8	0	8
main(String[])		0	0	0
c84_SmartWaterPurifier(double,		0	0	0
adjustFilterActivation()	100.0 %	4	0	4
updateTDS(double)	100.0 %	4	0	4

7. Z3 instruction coverage: 93.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	93	21,109	21,202
▼ src	0.4 %	93	21,109	21,202
▼ code	0.4 %	93	21,109	21,202
▼ c84_SmartWaterPurifier.java	93.9 %	93	6	99
▼ c84_SmartWaterPurifier	93.9 %	93	6	99
main(String[])	100.0 %	17	0	17
c84_SmartWaterPurifier(double,	100.0 %	15	0	15
adjustFilterActivation()	100.0 %	24	0	24
updateTDS(double)	100.0 %	37	0	37

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c84_SmartWaterPurifier.java	100.0 %	8	0	8
▼ c84_SmartWaterPurifier	100.0 %	8	0	8
main(String[])		0	0	0
c84_SmartWaterPurifier(double,		0	0	0
adjustFilterActivation()	100.0 %	4	0	4
updateTDS(double)	100.0 %	4	0	4

9. Unimproved V-method instruction coverage: 93.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	93	21,109	21,202
▼ src	0.4 %	93	21,109	21,202
▼ code	0.4 %	93	21,109	21,202
▼ c84_SmartWaterPurifier.java	93.9 %	93	6	99
▼ c84_SmartWaterPurifier	93.9 %	93	6	99
main(String[])	100.0 %	17	0	17
c84_SmartWaterPurifier(double,	100.0 %	15	0	15
adjustFilterActivation()	100.0 %	24	0	24
updateTDS(double)	100.0 %	37	0	37